Network Basics

Level 0–1 IT Basics - 04 Networking

Security Level:

HUAWEI

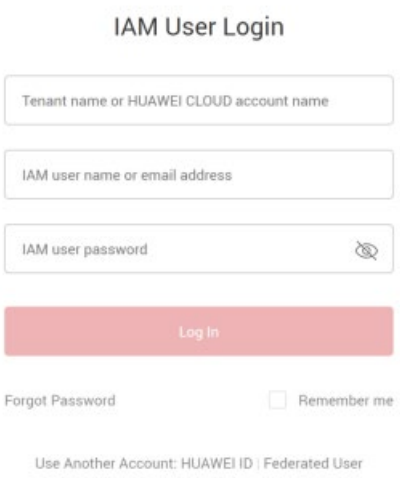
Identity and Access Management

Identity and Access Management (IAM)

- Basic functions:
  - Identity authentication
  - Access management
- Refined permissions management
- Huawei Cloud service authorization
- Identity federation with third-party identity providers

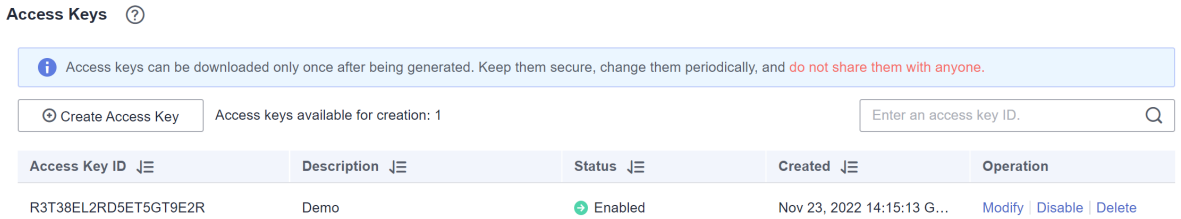
Identity Authentication

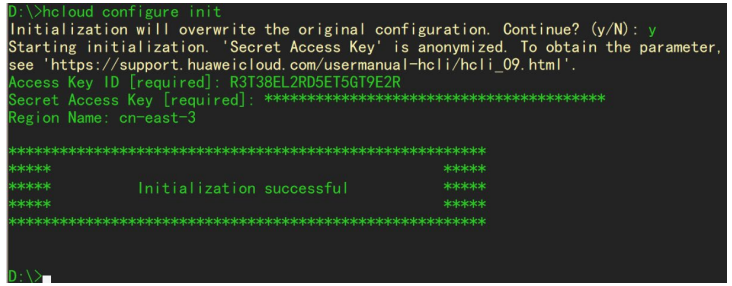
Method1: IAM user login



- Open the Huawei Cloud console login page.
- Use the IAM username and password to log in.
- Perform fine-grained permissions management on the IAM console

Method2: AK/SK for API access





- Use an access key (AK/SK) to verify your identity.
- Each IAM user can create two pairs of access keys.
- An AK/SK pair is used only for API access.

IAM User Groups

- An IAM user group is a collection of IAM users.
- An IAM user can belong to different IAM user groups.
- User groups make it easier to manage permissions for users.

IAM Agencies

IAM agency

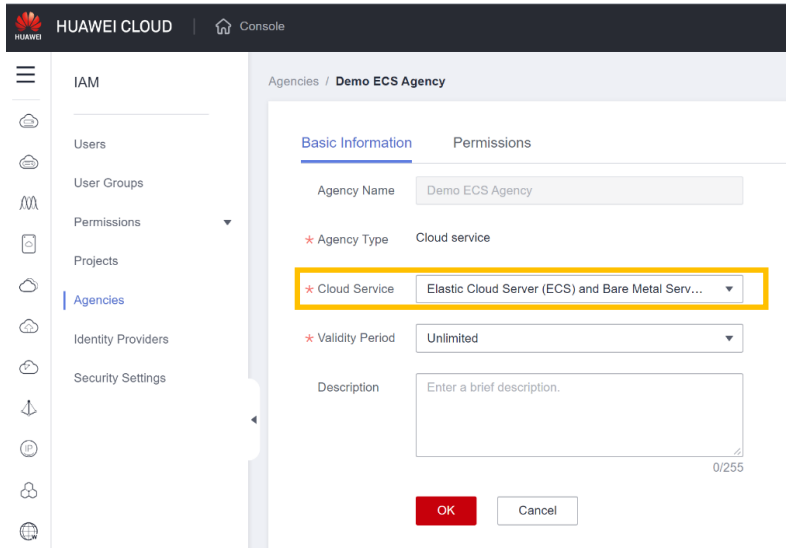
- Agencies do not have static credentials.
- Agencies get permissions through policy attachment.
- Agencies enable you to delegate permissions to:
  - Huawei Cloud services
  - Other Huawei Cloud accounts
  - Third-party identity providers

Associate Agency to ECS

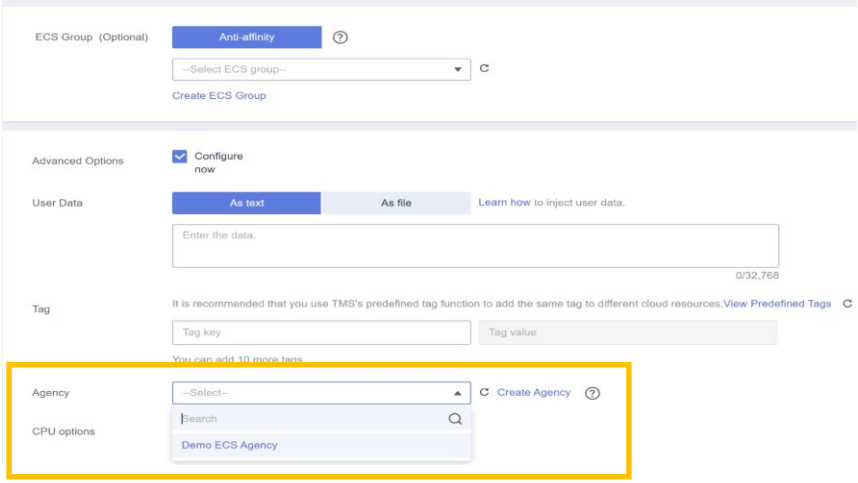
Obtain agency permissions to access OBS

OBS

On the **Permissions** tab, grant the agency permissions of calling APIs to access OBS.



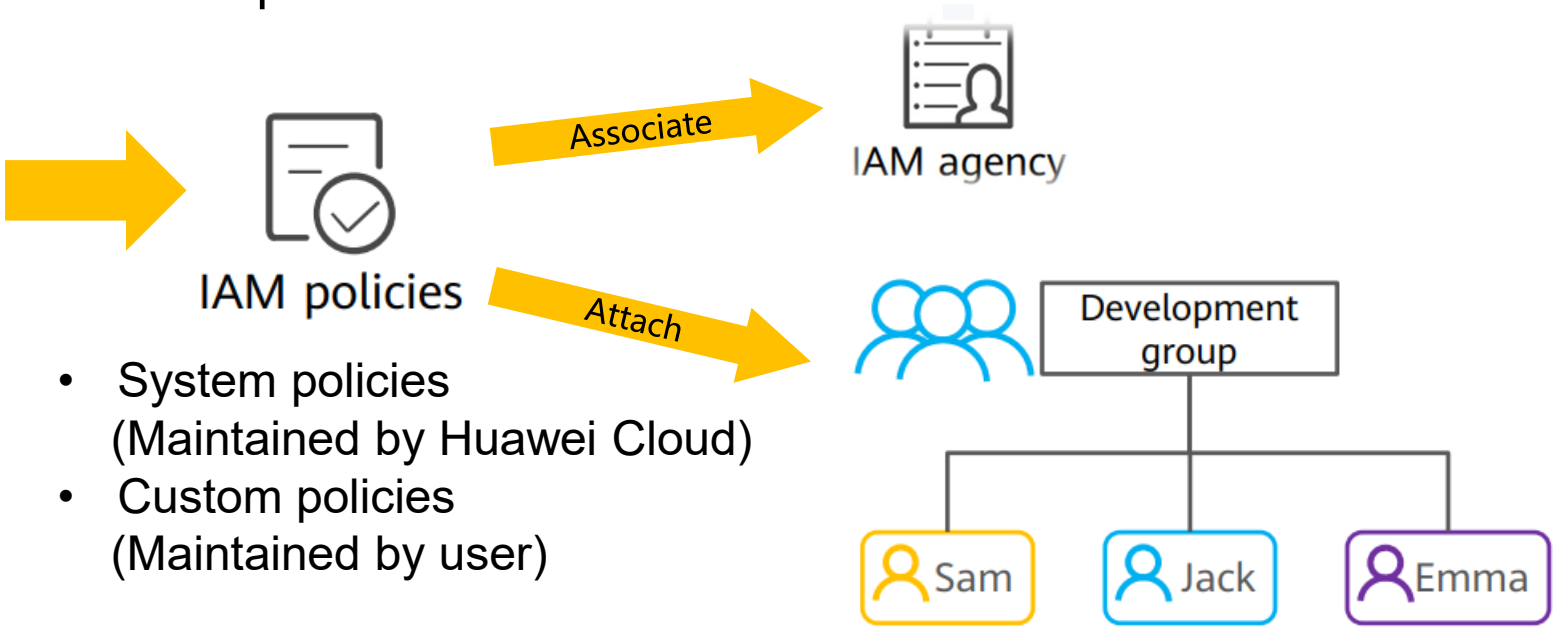
Associating the Created Agency with the ECS



IAM Permissions

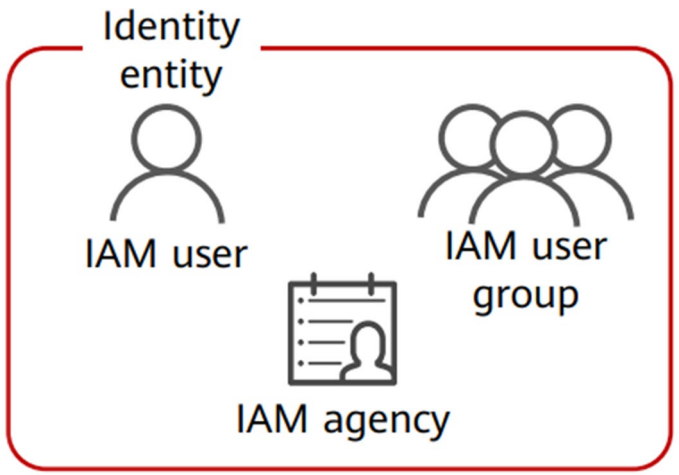
```
{
    "Version": "1.1",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "obs:bucket:ListBucket",
                "obs:bucket:Get*"
            ],
            "Resource": [
                "obs:*:*:bucket:*"
            ],
            "Condition": {
                "StringEndWithIfExists": {
                    "g:UserName":
["specialCharacter"]
                },
                "Bool": {"g:MFAPresent":
["true"]}
            }
        }
    ]
}
```

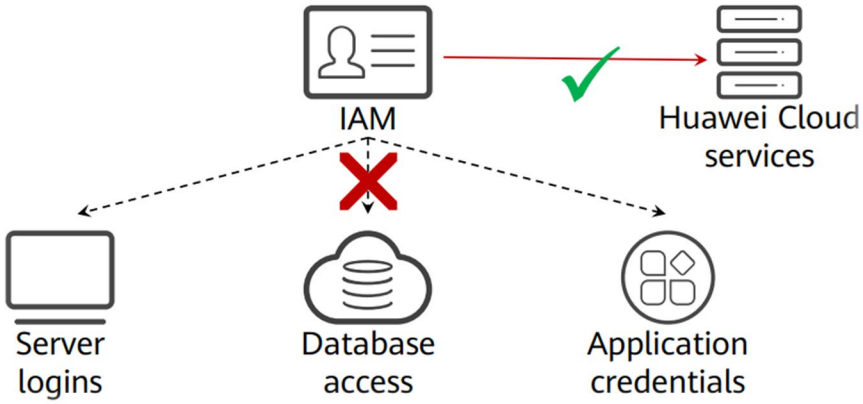
- IAM permissions are defined in JSON documents.
- JSON documents are encapsulated into policies for repeated use.



- System policies
  (Maintained by Huawei Cloud)
- Custom policies
  (Maintained by user)

IAM Review





- Identity authentication
  - Console
    User name and password
  - Programmatic access
    AK/SK
- Access management
  - Policies defined by JSON documents

Data Encryption Workshop (DEW)

Data Encryption Workshop (DEW)

- Secret management
- Database and server passwords can be stored in a centralized manner.
    - You no longer need to write them in your code or configurations.
- Seamless access with IAM agencies

Data Encryption Workshop (DEW)

Data **E**ncryption **W**orkshop (**DEW**) is a full-stack data encryption service in the cloud. It provides three sub-services: **Dedicated HSM**, **Key Management Service (KMS)** and **Key Pair Service (KPS)**. DEW protects keys by using hardware security modules (HSMs), and it is integrated with many other services on Huawei Cloud. Additionally, DEW enables you to develop customized encryption applications.

1 Dedicated HSM

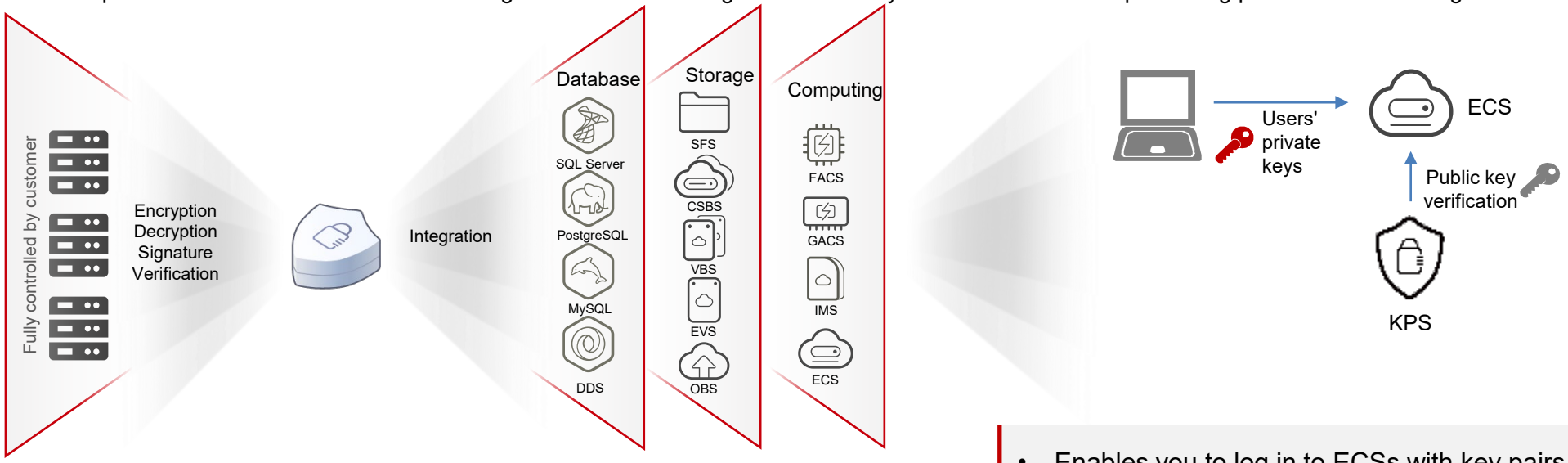
Dedicated HSM suits encryption scenarios requiring strict compliance.

2 KMS (Key Management Service)

Integrated with cloud services, KMS can manage and distribute a large number of keys.

3 KPS (Keypairs)

Implements login with private keys, preventing passwords from being cracked.



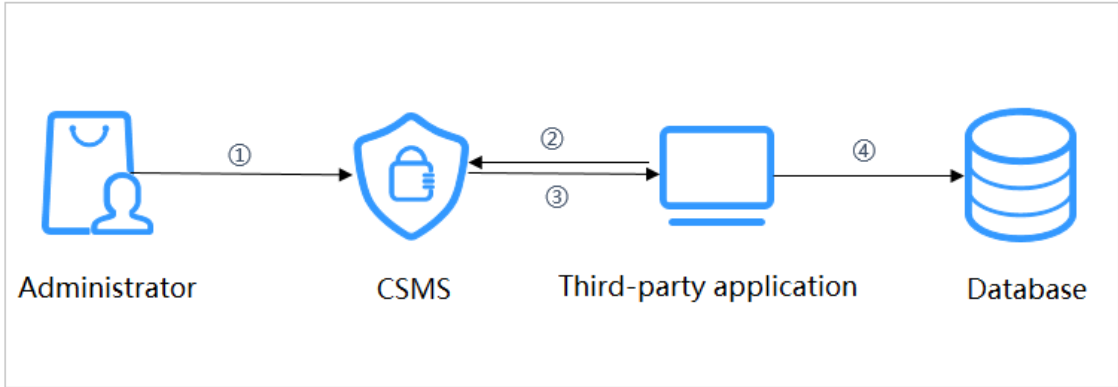
- Secondary development is required for using API requests to connect with applications.
- Dedicated HSMs ensure encryption performance.
- Supports CSCA algorithms, meeting compliance requirements.

- Integrated with basic storage services, requiring no development
- Pay-as-you-go, lower cost
- Root key protection, high security

- Enables you to log in to ECSs with key pairs securely and conveniently
- Private keys can be hosted rather than stored locally, avoiding loss
- Key pairs can be reset or replaced, providing higher security.

Cloud Secret Management Service - CSMS

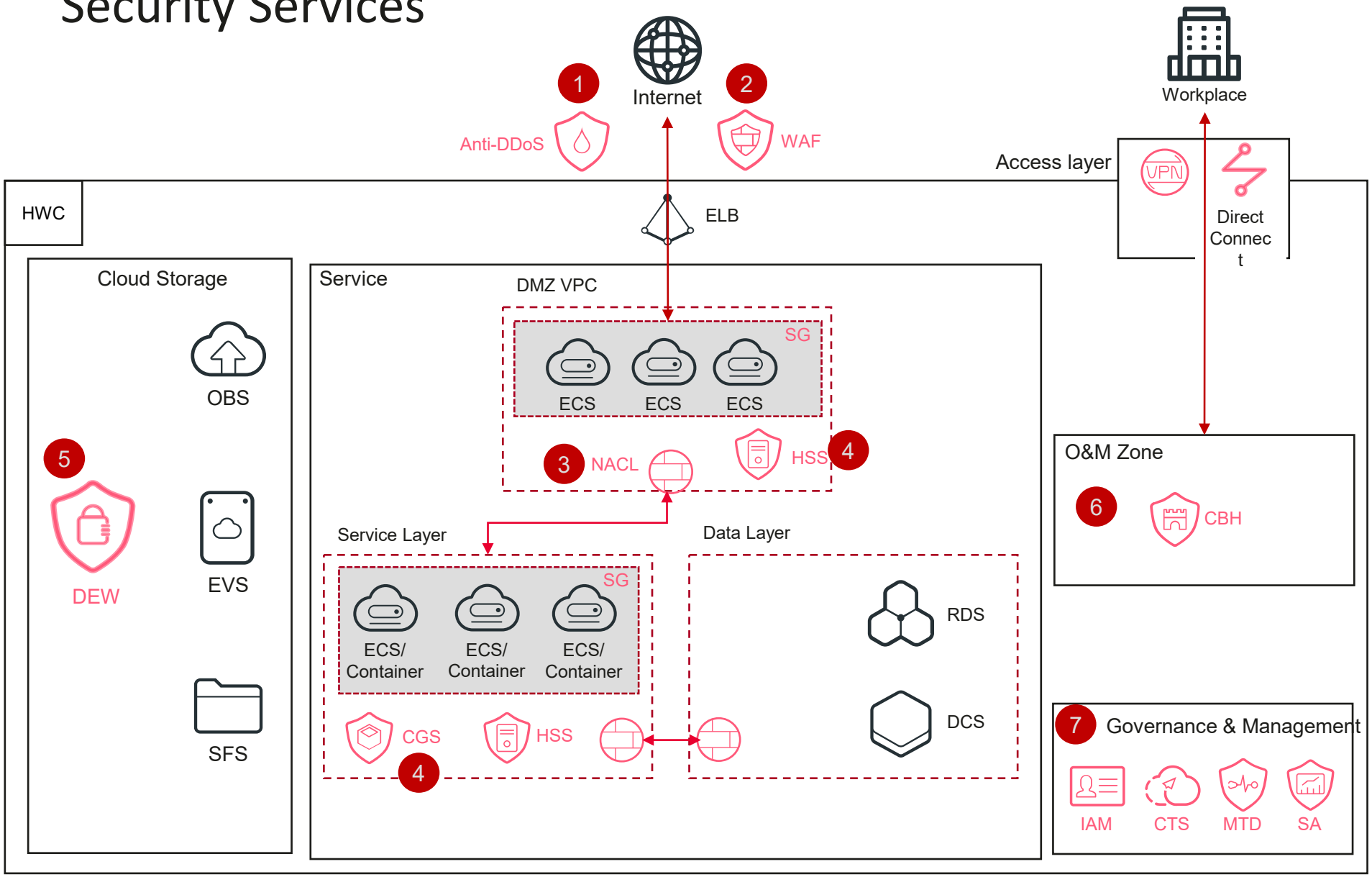
Cloud Secret Management Service (CSMS) helps you easily manage and **protect account information**, **tokens**, **certificates**, **SSH keys**, **API keys**, and **other sensitive data** throughout their lifecycles. You can create, retrieve, update, and delete secrets on the CSMS console or by using applications.



Scenarios

1. Create a secret on the console or via an API to store database information (such as the database address, port, and password).
2. Use an application to access the database. CSMS will query the secret you created.
3. CSMS retrieves and decrypts the credential ciphertext, and securely returns the information stored in the credential to the application through the credential management API.
4. The application obtains the decrypted plaintext secret and uses it to access the database.

Security Services



1 **AAD and Anti-DDoS:** Advanced Anti-DDoS and Traffic Cleaning (Layer 4)

2 **WAF:** Web Application Firewall (Layer 7)

3 **SG/NACL:** Security Group/ Network Access Control List

4 **HSS/CGS:** Host Security Service / Container Guard Service

5 **DEW:** Data Encryption Workshop

6 **CBH:** Cloud Bastion Host

7 **CTS/IAM/SA/MTD:** Cloud Trace Service/
Identity and Access Management/
Multi-Threat Detection/
Situation Awareness

Cloud Trace Service (CTS)

Cloud Trace Service (CTS)

- Manage **operation logs** on Huawei Cloud resources.
- Filter and query records from the last seven days.
- Analyze real-time LTS operation logs.
- Store logs in OBS for as long as needed.
- CTS itself is free of charge.
- Charges apply if you transfer traces to
    - OBS, LTS, and DEW.

Application Orchestration Service(AOS)

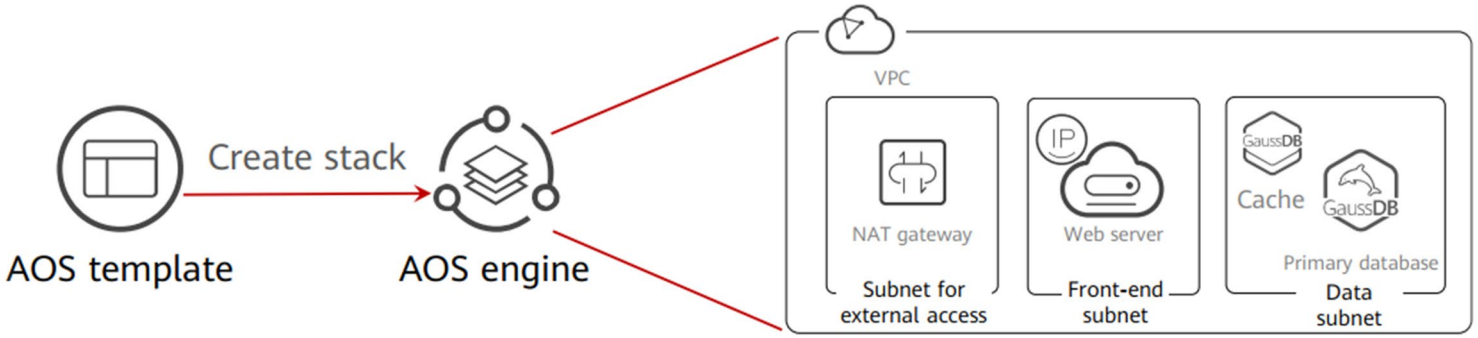
Application Orchestration Service
(AOS)

- Creates a template using YAML or JSON text.
- Manages application resources together.
  - Create, update, and delete actions.
- Suitable for multi-environment scenarios.
  - Agile updates
  - Consistent test, development, and production
  - Global, multi-region deployment

AOS Template

```
node_templates:
my-vpc:
type: HuaweiCloud.VPC.VPC
properties:
cidr: 192.168.0.0/16
my-subnet:
type: HuaweiCloud.VPC.Subnet
properties:
availabilityZone: cn-east-3a
cidr: 192.168.1.0/24
```

AOS starts all components in order and dependency as specified by the template.



AOS

- Create a template on the console or by calling APIs.
- Change templates using code management tools for **infrastructure as code**.
- AOS is **free of charge**, but other services it uses are not.
- Download **many public templates** on the console for reuse.
- When **sharing templates** with others, be aware of environment differences.